P.3 The Cartesian Plane and Graphs of Equations

Name _____ Period ____ Date _____

Pre-Calculus P.3 · Student sheet

I can…

- use the Distance and Midpoint Formulas.
- find intercepts and test a graph for symmetry.
- write and read the standard equation of a circle.

Vocabulary · fill in the missing word

Intercepts — the points where a graph _____ the x-axis or the y-axis

Symmetry — a graph is symmetric when a _____ maps it onto itself

Standard form of a circle — $(x - h)^2 + (y - k)^2 =$ _____

Part 1 · Fold for symmetry

Graph $y = x^2 - 2$.

x	−3	−2	−1	0	1	2	3
y							

Part 2 · Notes and examples (fill in each note, then work the example)

Distance: d = _____. **Midpoint:** $((x_1 + x_2)/2, (y_1 + y_2)/2)$

Example 1

Find the distance and the midpoint between (−2, 1) and (3, 4).

Example 2

Show that (2, 1), (4, 0), and (5, 7) form a right triangle.

Intercepts: for x-intercepts let y = _____; for y-intercepts let x = 0.

Example 3

Find the intercepts of $y = x^2 - 4x$.

Symmetry tests: replace y with −y (x-axis), x with −x (y-axis), or both (the _____).

Example 4

Test for symmetry.

a. $y = x^3 - x$

b. $x - y^2 = 1$

c. $y = x^2 + 2$

Circle: $(x - h)^2 + (y - k)^2 =$ _____

Example 5

Write the standard equation of the circle with center $(-1, 2)$ that passes through $(3, 4)$.

Part 3 · Partner practice

Partner 1

Find the distance and midpoint between $(-4, 10)$ and $(4, -5)$.

Partner 2

Write the equation of the circle with center $(2, -3)$ and radius 4.